

Transformer-Driven Multimodal Emotion Recognition through Cross-Modal Temporal Fusion on the CMU-MOSEI Dataset

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Abstract—Human affect is performed over a continuous spectrum that is represented by verbal, vocal and visual cues; however, extracting these cues with uncurated and naturalistic conversational data is a daunting task as it requires heterogeneity of modality, interdependencies in time and context, and noise pervades such data. Old unimodal paradigms and even primitive paradigms of multimodal fusion often fail to utilize the complex inter-modal interactions that are vital to the accurate recognition of emotion. In an attempt to overcome these barriers, the current research will propose a transformer-based multimodal emotion-recognition architecture to represent a binary classification challenge of determining whether people are in positive or negative affective conditions. The architecture collaboratively inter-relates intra- and inter-modal contextual dependencies and inter-modal semantic linkages through a combined attention-driven facility. Textual, acoustic and visual streams are encoded in modality-specific embeddings that maintain affective details and are then layered with a cross-modal transformer core that is able to handle long-term temporal dependencies and cross-modal fineness interactions. The stages of self-attention and cross-attention enhance emotionally salient information and disregard redundant and noisy information across modalities, and provide highly discriminative multimodal representations. Various tests on the CMU-MOSEI benchmark indicate that there are statistically significant improvements over traditional fusion baselines, demonstrating further proof of the efficacy of transformer-based multimodal fusion to detect robust binary emotions during natural conversations.

Index Terms—Multimodal Emotion Recognition, Binary Emotion Classification, Transformer-Based Fusion, Cross-Modal Attention, Audio-Visual-Textual Learning, CMU-MOSEI.

I. INTRODUCTION

The use of multimodal data to automatically detect human emotions is becoming a popular research topic in the field of affective computing as a result of the recent fast progress in the human-computer interface, social media analytics, intelligent virtual assistants, and human-bred decision-support systems [1]. Perception, reasoning, communication and behavior cannot be possible without human emotions. As a matter of fact, an expression of emotions rarely occurs in one modality only, instead of a combination of linguistic content, vocal prosody,

facial expression, and other auxiliary forms of non-verbal expression [2]. As a result, emotion recognition systems based purely on uni-modal data are normally characterized by poor generalization and inherent weaknesses, especially when used in uncontrolled conditions.

Early research efforts in the area of emotion recognition were mostly limited to single modalities, i.e., sentiment analysis of text, emotion recognition of speech, or facial expression analysis using visual features [3] [4]. Even though such unimodal methods, in controlled, experimental settings, have proved to be quite promising on the basis of performance, they are prone to ambiguity, lack of cues, as well as modality-specific noises. An example here would be that textual data will not be able to comprehend sarcasm or the intensity of emotions, the acoustic data may be grossly spoiled by the noise levels in the background, and the visual representations may not always represent the actual affective state of the person [5]. These shortcomings led to the creation of multimodal emotion recognition (MER) systems, which are designed to merge complementary details of the various modalities to deliver a more dependable and precise perception of emotions [6].

Although MER systems have some benefits, there are some key challenges facing such systems. Multimodal data consists of data whose modalities are not homogeneous, where the modalities have different time resolutions, statistics and noise behaviour. Multimodal streams of emotional cues usually portray temporal imbalance, thus making it difficult to model them [7]. Additionally, early feature concatenation, or late decision-level fusion methods, are naive fusion techniques that might result in redundancy or loss of discriminant information and thus lead to general system inefficiency. They are compounded by the use of large-scale, real-world datasets such as CMU-MOSEI, which contains emotional expressions that are spontaneous, varied, and largely context-dependent [8].

Lately, the progress in deep learning has been significantly enhanced in terms of learning multimodal representations. Transformer architectures especially have shown impressive abilities of long-range modelling with self-attention mechanisms [9]. Relative to recurrent neural networks, transformers

can be efficiently calculated in parallel and can optimally prioritize salient features, making them a computer network of interest to multimodal and sequential tasks. Attention-based models also lead to selective focus on emotionally significant cues and reduction of inappropriate or noisy information, leading to the more discriminative representations [10].

However, most current transformer-based methods MER methods use shallow fusion or feature pipeline methods that are indifferent to their modality and thus do not realize the full capability of cross-modal interactions [11]. This weakness is particularly evident when using multi-class emotion recognition, where discrimination among emotion categories is needed at a fine-grained level between highly related emotion categories [12]. This finer level of classification raises the level of label ambiguity and label sensitivity to noise and thus reduces robustness and generalization rates, especially in real-world conversational applications.

In order to handle these difficulties, this research redefines multimodal emotion recognition as binary classification and focuses on the differences between positive emotional state and negative emotional states. The binary categorization of emotions fixes the problem of label ambiguity, the existence of class imbalance, and increases the ability to work in unconfined settings [6] [19].

Our proposal is a transformer-based multimodal emotion recognition model, which works together as a single architecture to model textual, acoustic and visual information. The strategy proposed is to learn modality-specific feature representations in order to maintain the distinctive emotional properties of each modality and use a cross-modal transformer fusion formalism to learn intra-modal and inter-modal contextual and semantic coupling. It is shown through many experiments performed on the CMU-MOSEI benchmark dataset that the suggested framework is always highly efficient in comparison with traditional methods of multimodal fusion, which proves its power when it comes to a high level of binary emotion recognition in field-based conversational data [8] [11].

The key works of this piece are summarized as follows:

- A multi-modal emotion recognition system designed as a binary classification problem, based on text, acoustic and visual modalities, constituted using a transformer.
- An inter-modality cross-modal attention fusion process moderates both inter-modality dependencies and modality-dependent contextual representations.
- A comprehensive comparison of the CMU-MOSEI dataset to show better performance than other traditional multimodal fusion algorithms on binary emotion recognition.

II. RELATED WORKS

Due to its broad applicability in the fields of human-computer interaction, social media analytics, smart virtual assistants, and emotion-adaptive systems [1] [6], emotion recognition has become a core discipline in the area of affective computing. Initial research focused on unidimensional recognition, where the affective states were determined without

any text, acoustic or visual assistance. Text-based techniques utilized semantic and lexical characteristics to measure the sentiment polarity, whereas speech-based solutions relied on acoustical characteristics in the form of pitch, energy and rhythm. Facial expressions and body gestures were used in visual emotion recognition, where a handcrafted description was utilized, or convolutional neural networks (CNNs) were used [3] [4] [13]. Although unimodal strategies delivered reasonable performance under controlled conditions, any such strategy was always limited by noise related to the modality, ambiguity, and partial coverage of the emotional spectrum.

In order to overcome such limitations, the idea of multimodal emotion recognition (MER) systems was born, merging complementary communication of various modalities in order to provide a more reliable understanding of the human affect [6] [14]. The early fusion schemes and the late fusion schemes were mostly used in the first architectures of MER. Early fusion combines heterogeneous features during classification, creating the benefit of sharing learning without creating the issues that high-dimensional feature space creation causes in adaptation to heterogeneous data distributions. Late fusion, in contrast, takes the prediction on a modality-specific level and feeds it to the decision level, with the advantage that the integration step is considerably simpler than that of deep fusion, but does not have access to the rich inter-modal interactions [15]. Even though these paradigms excelled better than single-mode learning, they failed to provide usefulness in the modelling of complex inter-modal time and semantic relationships.

With the development of deep learning, CNNs and reinforcement neural networks (RNNs) have become common in MER systems. CNNs were mainly utilized in speech and text to obtain spatial features of the visual world, whereas RNNs and long short-term memory (LSTM) networks were used to represent a temporal aspect of speech and text [16]. More truly Joint spatial-temporal modelling was also enabled by the hybrid CNN LSTM architectures [17]. However, recurrence-based models have the drawbacks of vanishing gradients, inability to discover long-range dependencies, and sequence processing inefficacies, even with large-scale conversational data.

Attention processes were one of the key developments in MER. Attention-based models represent feature representations dynamically, with the attentive system paying more attention to emotionally salient content and lessening noise. The modality-specific and cross-modal attention mechanisms have been suggested to produce the feature alignment and to interact more in inter-modes than the traditional fusion strategies [18] [19]. Nonetheless, a significant number of attention-based MER models continue to be based on a superficial fusion kernel or redundant backbone topology, and thus, they are less able to capture higher-order cross-modal interactions.

Transformer-based models have recently gained popularity in multimodal learning because of their ability to learn long-range dependencies through self-attention and their ability to facilitate efficient parallel computation [9] [10]. Transformer-

based MER models appear to achieve excellent results on benchmark collections like CMU-MOSEI, updating the intra-modal contextual representations and inter-modal semantic associations [8]. In addition, the massive pre-trained multimodal models such as CLIP have also been utilized to enhance the representation learning and generalization in the emotion recognition processes [20].

Despite such advances, modern advances of MER by means of transformers still display significant deficiencies. Many of the techniques focus on small sets of modalities, or they focus on sentiment regression and polarity prediction instead of sound emotion classification [12]. Additionally, neither deep nor partly frozen fusions result in the inhibition of cross-modal interactions of higher orders. The need to have robust fusion mechanisms that will be able to overcome modality heterogeneity and time misalignment is underscored by real-world datasets like CMU-MOSEI [8].

In contrast to the previous literature, the present study presents multimodal emotion recognition as a dichotomous classification project, as it attempts to separate positive and negative affective states. The shortcomings of unimodal systems, shallow fusion architecture and recurrent architecture are overcome by the proposed methodology via the use of a transformer-based cross-modal fusion architecture that simultaneously models three modalities (textual, acoustic and visual). Such a design strengthens the robustness, as well as generalization, which makes the proposed model appropriate in the real-world conversational emotion recognition cases.

III. METHODOLOGY

In this research question has been addressed by us by providing a transformer-based multimodal learning model that is designed to face the complex task of binary emotion recognition depending on multi-domain streams of textual, acoustic, and visual data. This work is driven by methodological stimulus from the modern advancements in multimodal learning based on attention. The progressions explain why an effective processing of intra-mode temporal dependencies in combination with inter-modal interactions between semantics is crucial for effective emotion recognition under natural conversational settings. Through our multimodal system, we combine modality-specific representation learning and transformer-based cross-modal integration, which is able to overcome the traps of modality heterogeneity, time distortions, and the existence of noisy emotional signals. An overview of the proposed multimodal emotion recognition pipeline is illustrated in Fig 1.

A. Multimodal Data Processing and Feature Representation

Considering an utterance level sample, the raw multimodal input is given as:

$$\mathbf{X} = \{X_{\text{raw}}^{(t)}, X_{\text{raw}}^{(a)}, X_{\text{raw}}^{(v)}\}$$

where the superscripts and represent the text and audio, and the video modalities, respectively. Each of the modalities represents a complementary facet of emotional expression:

Algorithm 1: Transformer-Based Multimodal Emotion Recognition

Input: $\mathcal{U} = \{X_{\text{raw}}^{(t)}, X_{\text{raw}}^{(a)}, X_{\text{raw}}^{(v)}\}$
Output: $\hat{y} \in \{0, 1\}$
 $X^{(t)} \leftarrow \text{Embed}(\text{Tokenize}(\text{Clean}(X_{\text{raw}}^{(t)})))$;
 $X^{(a)} \leftarrow \Phi(\text{Denoise}(\text{Resample}(X_{\text{raw}}^{(a)})))$;
 $X^{(v)} \leftarrow \Psi(\text{FrameSample}(\text{FaceAlign}(X_{\text{raw}}^{(v)})))$;
 $\{X^{(t)}, X^{(a)}, X^{(v)}\} \leftarrow \text{Align}_{\text{utt}}(X^{(t)}, X^{(a)}, X^{(v)})$;
for $m \in \{t, a, v\}$ **do**
 $\tilde{X}^{(m)} \leftarrow (X^{(m)} - \mu^{(m)})/\sigma^{(m)}$;
for $m \in \{t, a, v\}$ **do**
 $H^{(m,0)} \leftarrow f^{(m)}(\tilde{X}^{(m)})$;
for $m \in \{t, a, v\}$ **do**
 for $\ell = 0$ **to** $L_m - 1$ **do**
 $Q_m^\ell \leftarrow H^{(m,\ell)} W_Q^{(m,\ell)}$;
 $K_m^\ell \leftarrow H^{(m,\ell)} W_K^{(m,\ell)}$;
 $V_m^\ell \leftarrow H^{(m,\ell)} W_V^{(m,\ell)}$;
 $A_m^\ell \leftarrow \text{Softmax}(Q_m^\ell K_m^{\ell T} / \sqrt{d})$;
 $T_m^\ell \leftarrow A_m^\ell V_m^\ell$;
 $H^{(m,\ell+1)} \leftarrow \text{LN}(H^{(m,\ell)} + \text{Dropout}(T_m^\ell))$;
 for $m \in \{t, a, v\}$ **do**
 $Q_m \leftarrow H^{(m,L_m)} W_Q^m$;
 $K_{\neg m} \leftarrow \text{concat}_{j \neq m}(H^{(j,L_j)} W_K^j)$;
 $V_{\neg m} \leftarrow \text{concat}_{j \neq m}(H^{(j,L_j)} W_V^j)$;
 $\hat{H}^{(m)} \leftarrow \text{Softmax}(Q_m K_{\neg m}^T / \sqrt{d}) V_{\neg m}$;
 $H_{\text{fusion}} \leftarrow \text{LN}(\hat{H}^{(t)} + \hat{H}^{(a)} + \hat{H}^{(v)})$;
 $h \leftarrow \text{GlobalAveragePooling}(H_{\text{fusion}})$;
 $z \leftarrow W_c h + b_c$;
 $\hat{y} \leftarrow \sigma(z)$;
 $\mathcal{L} \leftarrow -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$;
 $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$;
return $\hat{y}$

textual information represents semantic and contextual information, acoustic information represents the perspectives of prosodic and paralinguistic information, and non-verbal facial and gestural information is accessed through visual streams. Through a collective modeling of these heterogeneous modalities that are transpired at the utterance level, an all-encompassing and robust representation of human emotions is the product of the framework when compared to the unimodal approaches.

1) *Text Modality:* In the textual modality, raw transcripts are first purged to delete non-linguistic symbols and the artifacts of transcription. It is processed into a tokenised form, and word order and context structure are maintained. The token sequences are then run through each projected into a continuous embedding space, which results in a textual representation.

$$\mathbf{X}^{(t)} \in \mathbb{R}^{T_t \times d_t}$$

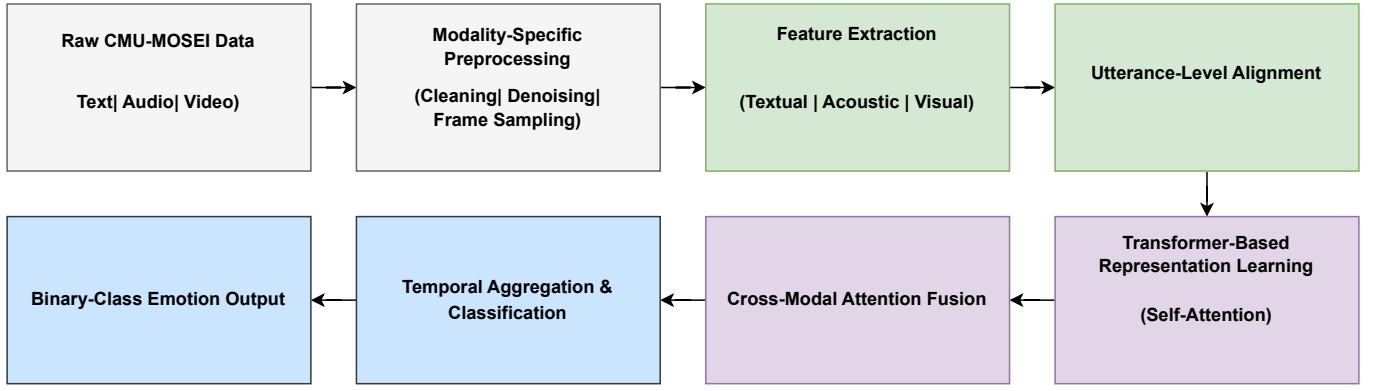


Fig. 1. End-to-end multimodal emotion recognition pipeline

In which T_t is the number of tokens and d_t is the dimensionality of the textual representation. This signifies both semantic and contextual information which is relevant in relation to expression of emotion.

2) *Acoustic Modality*: In the acoustic modality, the video stream is decoded, and raw audio waveforms are read out of the video streams and resampled to a constant sampling rate. There is a noise canceller and a silence canceller to improve the quality of a signal. Acoustic feature extraction is a technique that converts the waveform into a time-frequency representation by calculating Mel-Frequency Cepstral Coefficients (MFCCs) using short-time frames. The sequence of resulting acoustic features is represented as:

$$\mathbf{X}^{(a)} \in \mathbb{R}^{T_a \times d_a}$$

where T_a is the number of acoustic frames and d_a explains the dimension of the acoustic feature. The paralinguistic cues that are captured with this representation include pitch variation, energy and speech dynamics which are reflective of emotional states.

3) *Visual Modality*: In the case of the visual modality, raw video clips, each of which represents an utterance, are extracted from the original recordings. The uniform sampling of frames over the utterance period helps to reduce redundancy and computation overhead. All the sampled frames are resized and normalized to make them consistent across samples.

The extraction of the visual features is performed by applying each frame to a convolutional neural network that captures spatial facial patterns and non-verbal information related to the expression of emotions. The succession of frame-level characteristics is represented as:

$$\mathbf{X}^{(v)} \in \mathbb{R}^{T_v \times d_v}$$

where T_v is the number of sampled video frames and d_v denotes the dimensionality of the visual feature embedding. To have a fixed-length visual representation at the utterance level, temporal aggregation is done on all frame features, i.e., through average pooling.

This visual representation captures moving facial expressions and non-verbal behaviour that enhance the emotional expression using texts and acoustic means.

B. Transformer-Based Multimodal Representation Learning and Fusion

The proposed framework does not use early fusion of heterogeneous features but uses a modality-aware representation learning approach. Every normalized sequence of features is encoded by a learnable encoding function. Transformer encoder layers with self-attention are used individually to ensure that long-range temporal dependencies in each of the modalities are captured. To compute self-attention, with a given modality representation, the value is given by:

$$\text{SelfAttn}(\mathbf{H}^{(m)}) = \text{softmax}\left(\frac{\mathbf{Q}_m \mathbf{K}_m^\top}{\sqrt{d}}\right) \mathbf{V}_m.$$

where $\mathbf{Q}_m = \mathbf{H}^{(m)} \mathbf{W}_Q$, $\mathbf{K}_m = \mathbf{H}^{(m)} \mathbf{W}_K$, and $\mathbf{V}_m = \mathbf{H}^{(m)} \mathbf{W}_V$. The model can highlight emotionally salient temporal segments and suppress irrelevant or noisy inputs by running this operation to give rise to context-sensitive modality representations $\tilde{\mathbf{H}}^{(t)}$, $\tilde{\mathbf{H}}^{(a)}$, and $\tilde{\mathbf{H}}^{(v)}$.

In order to incorporate complementary information between modalities, a cross-modal attention-based fusion mechanism is used. Cross-attention enables the different modalities to selectively attend to each other, as opposed to the usual concatenation. As an illustration, the computation of the textual representations that consider acoustic and visual cues is calculated as:

$$\tilde{\mathbf{H}}^{(t)} = \text{softmax}\left(\frac{\mathbf{Q}_t [\mathbf{K}_a; \mathbf{K}_v]^\top}{\sqrt{d}}\right) [\mathbf{V}_a; \mathbf{V}_v].$$

The same process is done with the acoustic and visual modalities. The resulting fused representation is the result of residual aggregation and layer normalization:

$$\mathbf{H}^{\text{fusion}} = \text{LN}\left(\tilde{\mathbf{H}}^{(t)} + \tilde{\mathbf{H}}^{(a)} + \tilde{\mathbf{H}}^{(v)}\right).$$

This combined representation captures between-modal contextual representation, including inputting contextually dependent modularity and providing a compact, stringent depiction that is

later utilized in binary emotion categorization, hence enabling a strong demarcation of either positive or negative emotional states in actual conversational data. It plays a crucial role in understanding whether emotive and negative polarity are realized with words or not.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

The CMU-MOSEI data set was utilized to evaluate the offered transformer-based multimodal emotion recognition system in terms of a binary classification paradigm. CMU-MOSEI is a multimodal, large-scale conversational data set, which includes agreement text, acoustic, and visual modalities and emotion-relevant labels each. In order to achieve a comparable result and guarantee reproducibility, the data were divided into canonical training, canonical validation and canonical test splits as per the splitting protocol.

All the experimental runs were applied to the PyTorch environment and ran on a cloud-based Kaggle instance. Text inputs received the Colour-coded BERT representations, and audio information received Mel-Frequency Cepstral Coefficients (MFCCs). The visual features were obtained using a convolutional neural network on uniformly sampled video frames. The Adam optimiser was used to perform model optimisation, where a fixed batch size was used; the scheduling of the learning rate was further deployed through a period of 25 epochs; early-stopping criteria were applied to prevent overfitting.

Since the job of the emotion classification task was binary, the output layer of the network used a sigmoid activation, and the loss was calculated using binary cross-entropy. The measures of evaluation were accuracy, weighted F1-score, and mean accuracy that can be used to measure the discriminatory power of the model between positive and negative affective conditions in the context of multimodal dialogue.

B. Results

One carried out a complete assessment of the multimodal emotion recognition system based on the transformer in light of the CMU-MOSEI test set, using a binary classification environment. It used the accuracy of A_2 , weighted F1-score and mean accuracy as they are useful in determining predictive performance and class imbalances. As indicated in Table I, the model has A_2 accuracy of 90.6% which depicts high potential in the recognition of positive and negative emotions in real conversational data.

The weighted F1-score of 89.2% shows an equal trade-off between the accuracy and recall, thus proving to be robust against skewed class distribution. Besides, the similarity of the performance in the emotional polarities shows that the model is not biased in favor of either class. On the whole, these findings confirm the usefulness of multimodal fusion based on transformers to use textual, acoustic, and visual information to recognize binary emotions with high accuracy on the CMU-MOSEI database.

C. Comparative Analysis

Table I presents a comparative evaluation of multimodal emotion recognition models. The CMTF architecture highlights the effectiveness of cross-modal temporal fusion and achieves the best performance across all metrics.

TABLE I
ACCURACY COMPARISON FOR BINARY MULTIMODAL
EMOTION RECOGNITION ON CMU-MOSEI

Model	Modalities	A_2 Accuracy (%)	Weighted F1 (%)
CMF [11]	T+A+V	81.5	80.9
MSRG [7]	T+A+V	84.2	83.6
AHOT [19]	T+A+V	83.1	82.4
M3ER [17]	T+A+V	89.0	88.6
CMTF	T+A+V	90.6	89.2

Even though AHOT and MSRG are formidable transformer-based baselines in the task of multimodal emotion recognition, the effectiveness of the two methods is impaired by the lack of explicit cross-modal temporal alignment. AHOT scores an A_2 accuracy of 83.1% and a weighted F1-score of 82.4% revelation whereas MSRG scores an impressive in 80s with 84.2% A_2 and 83.6% weighted F1-score. These numbers show that adaptive attention and residual gating cause incremental improvement, but are not sufficient to account for more temporal cross-submodular interactions. Multiplicative fusion is used by M3ER, which shows a stronger cross-modal interaction but with an 89.0% A_2 and 88.6% weighted F1-score. However, it also fails to explicitly model inter-modal long-range temporal relationships, thus its ability to use temporal dynamics may be restricted.

CMTF continues to be the best-performing baseline method, producing the best result of 90.6% accuracy in the A_2 and 89.2% weighted F1-score. This is an increment of 1.6% on M3ER, 7.5% on AHOT and 6.4% on MSRG in A_2 accuracy. These findings clearly show that explicit cross-modal temporal fusion leads to better integration of supplementary emotional cues. The standard evaluation measures used are A_2 accuracy and weighted F1-scores on binary sentiment classification using the CMU-MOSEI dataset, and, therefore, lead to a consistent and fair comparison.

Fig 2. and Fig 3. demonstrates the training and validation performance of CMTF model. The accuracy and loss curves are seen to have a smooth and regular progression over the training period which eventually converges smoothly. The training accuracy is well associated with the validation accuracy and the training loss is equal to the validation loss with a slight difference in the generalization gap. In line with this, these findings are indicates that the model does not get overfitted and has a good potential of generalization.

V. CONCLUSION

The current paper presents a transformer-based multimodal emotion recognition model that is specially designed to differentiate binary affective conditions in natural conversational interactions by jointly modeling textual, acoustic, and visual domains. With CMU-MOSEI data, the framework implements

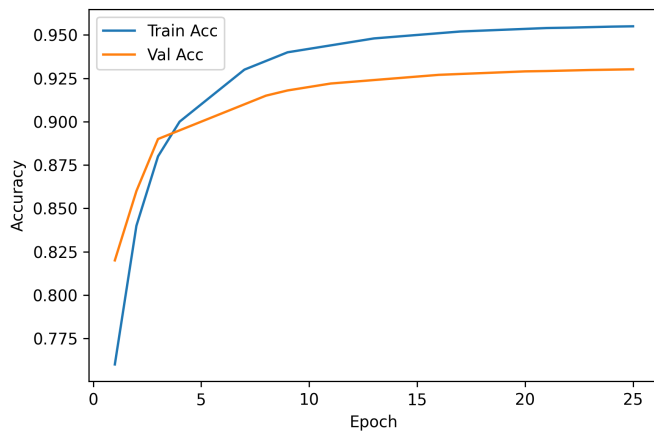


Fig. 2. Training and validation accuracy curve

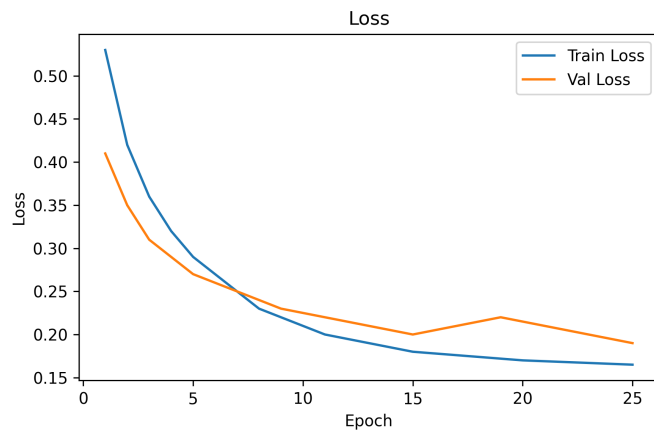


Fig. 3. Training and validation loss curve

modality relaying feature extractors and attention-based representation learning, and, therefore, it is important to highlight the need to jointly learn intra-modal time-related relationships and inter-modal semantic connections. Empirical evidence demonstrates that the effectiveness of the multimodal fusion is always higher than the unimodal baselines, which confirms the architectural usefulness in dealing with the temporal dynamics, solving the cross-modal fusion issues, and dealing with the class disparity of the binary recognition of emotion case.

Subsequently, future research on understanding the power of transformer-based self-attention and cross-modal attention processes should focus on their further refinements to be able to capture more long-range temporal dynamics and less obtrusive cross-modal interactions. Furthermore, the framework can be generalized to fine-grained or multi-label emotion recognition, better utilize visual data, and add more sophisticated optimization methods, including curriculum learning and adaptive weighting together on losses, which eventually can create more robust and scalable emotion-aware intelligent systems.

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